



Pharm*Achieve*

INFECTIVE ENDOCARDITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACC American College of Cardiology

AHA American Heart Association

AMG Aminoglycoside

[C] Concentration

CHD Congenital Heart Disease

GU Genitourinary

HD Heart Disease

ICH Intracranial Hemorrhage

IE Infective Endocarditis

MSSA Methicillin-Sensitive *S. aureus*

MRSA Methicillin-Resistant *S. aureus*

MIC Minimum Inhibitory Concentration

PWID Persons Who Inject Drugs

SHD Structural Heart Disease

TEE Transesophageal Echocardiography

TTE Transthoracic Echocardiography

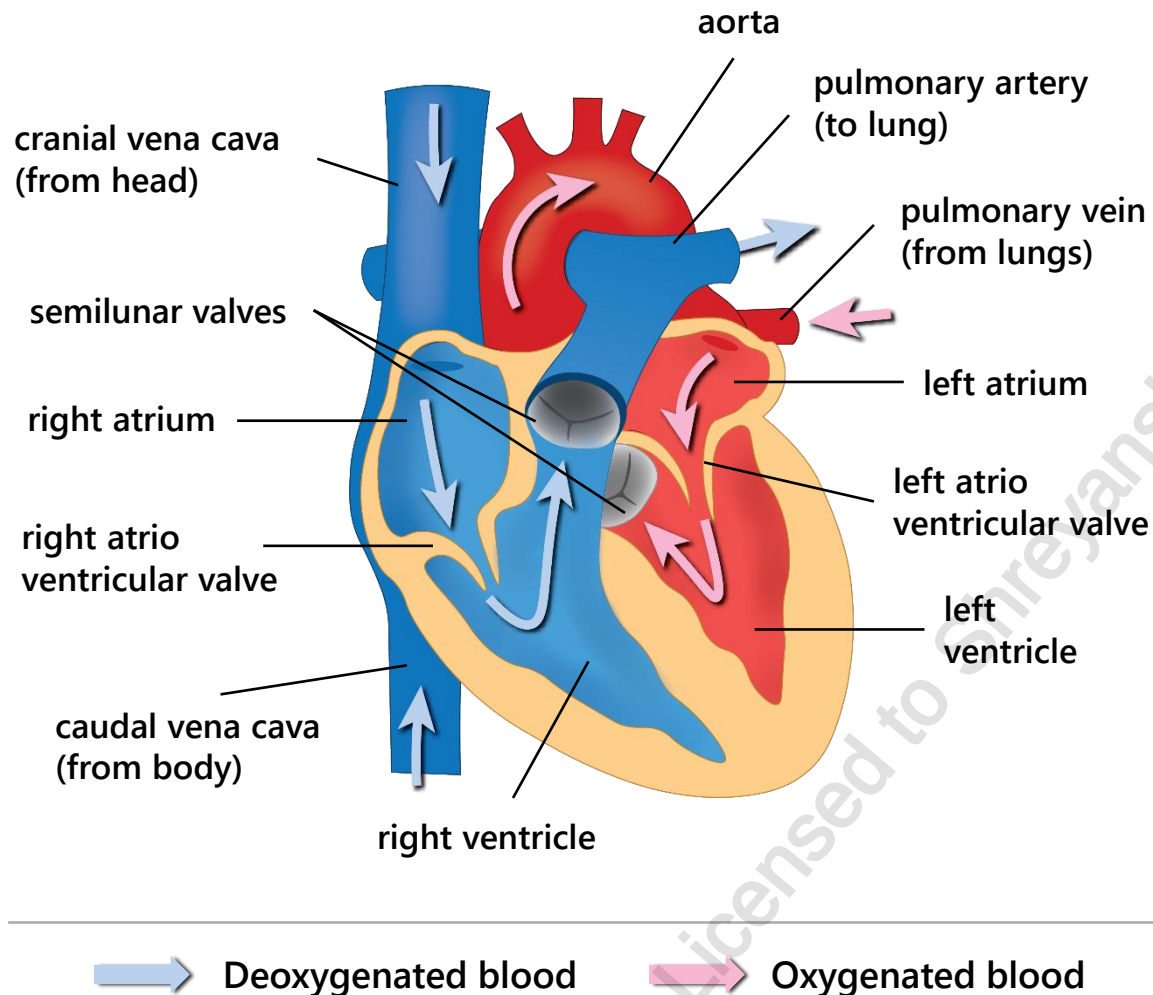
VGS Viridans Group Streptococci

CASE #1

AJ, a 45-year-old male with a history of injection drug use, presents to the emergency department with a 5-day history of fever, chills, and right-sided chest pain that is worsened by inspiration. He is tachycardic and febrile, with a new systolic murmur best heard along the left lower sternal border. Blood cultures grow *Staphylococcus aureus*, and transthoracic echocardiography reveals a vegetation on the tricuspid valve. He is diagnosed with right-sided infective endocarditis, likely related to intravenous drug use. Pulmonary septic emboli are noted on CT chest.

ANATOMY OF THE HEART

FYI



- There are 4 valves: aortic, tricuspid, pulmonary, mitral (bicuspid)
- Direction of blood flow through the valves is forward
- Opening and closing of valves is controlled by the blood pressure in each chamber
- Aortic, tricuspid, and pulmonary valves contain 3 leaflets, while mitral valve contains only 2

DEFINITIONS

Endocarditis: inflammation of endocardium (inner layer of the heart) usually involving the valves

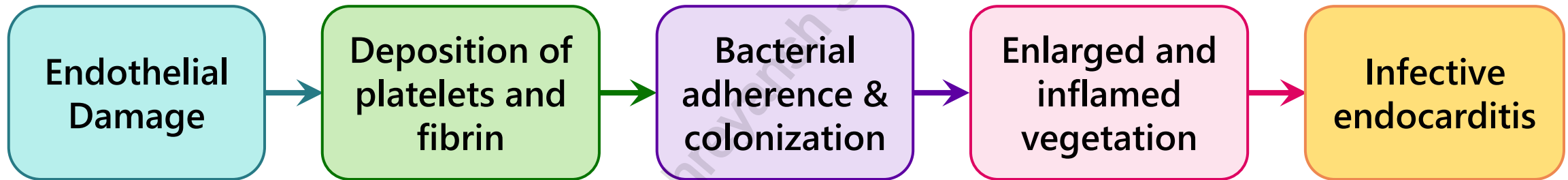
- Endocarditis lesions are known as vegetations
- Based on the source of inflammation, endocarditis is classified into two groups:
 - Infective endocarditis (IE)
 - Non-infective endocarditis

Vegetation: abnormal growth often associated with endocarditis

- Vegetations are composed of fibrin clots, platelets, and microorganisms

PATHOGENESIS

FYI



CLINICAL PRESENTATION

- Fever
- Malaise, general fatigue
- Nausea/vomiting, anorexia
- Night sweats
- Weight loss
- Dyspnea, pleuritic chest pain
- Headache, abdominal/back pain
- Sepsis (usually in acute bacterial endocarditis)
- Acute heart failure
- Cardiac murmur: due to valvular destruction and regurgitation
- Physical stigmata: splinter hemorrhages, Janeway lesions, Osler nodes, petechiae (buccal, mucosal, conjunctival), Roth spots, loss of visual acuity, pulmonary edema, neurologic deficits, splenomegaly

COMPLICATIONS

Complications: if left untreated, infective endocarditis is associated with 100% mortality

Embolic

- Systemic embolization most common in left-sided infective endocarditis
- Pulmonary embolism in right-sided lesions
- Causing stroke, paralysis, blindness, acute myocardial infarction, ischemia of extremities

Cardiac

- Valvular destruction leading to heart failure
- Perivalvular and metastatic abscesses
- Conduction abnormalities and heart block

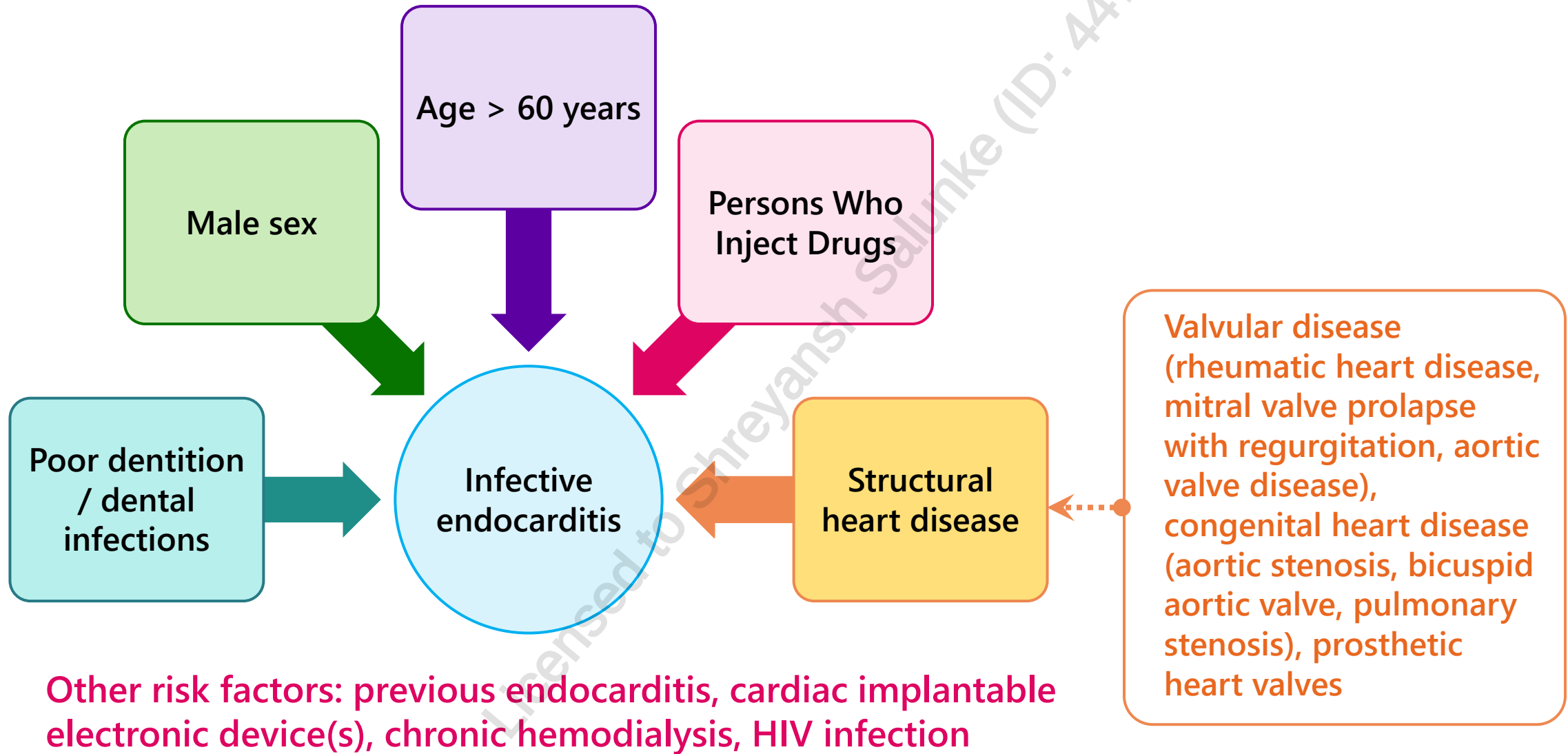
Immunologic

- Glomerulonephritis
- Roth spots
- Osler's nodes

Infectious

- Vertebral osteomyelitis and septic arthritis
- Meningitis
- Pulmonary infections: bacterial pneumonia, lung abscess, pleural effusion, pneumothorax

RISK FACTORS



ETIOLOGY

Gram-positive bacteria	Gram-negative bacteria	Other
40%: Staphylococci <i>Staphylococcus aureus</i>, MSSA, MRSA - Coagulase-negative staphylococci (e.g. <i>S. epidermidis</i>) 20%: Streptococci (Viridans streptococci - VGS, <i>S. bovis</i>) 10%: Enterococci (<i>E. faecalis</i>, <i>E. faecium</i>)	1-2%: HACEK organisms (includes: <i>Haemophilus parainfluenzae</i> , <i>Aggregatibacter</i> , <i>Cardiobacterium hominis</i> , <i>Eikenella corrodens</i> and <i>Kingella</i>)	8%: Difficult-to-grow pathogens - Often culture negative (e.g. <i>Coxiella burnetii</i>, <i>Bartonella</i> spp., <i>Chlamydia</i>) - Rarely: fungal (e.g. <i>Candida</i> spp)

3 groups of organisms causing the majority of IE cases:

<i>Staphylococci</i> (40%)	<i>Streptococci</i> (20%)	<i>Enterococci</i> (10%)
- Common cause of health-care associated IE <i>Staphylococcus aureus</i> is the most common causative agent of IE in the world and also specifically in IVDUs - Coagulase-negative staphylococci are less virulent and are more involved in prosthetic valve endocarditis	Common cause of community-acquired native valve IE. Likely source is odontogenic	May be seen in patients with recent history of genitourinary/obstetric procedures or GI malignancy

DIAGNOSIS

FYI

- The modified Duke criteria is the gold standard for diagnosis of IE
- A series of major and minor criteria based on clinical symptoms and investigations (e.g. blood culture, positive ECHO, fever $>38^{\circ}\text{C}$)

Definite IE: Pathological criteria + clinical criteria (2 major OR 1 major + 3 minor OR 5 minor)

Possible IE: 1 major + 1 minor criterion OR 3 minor criteria

Rejected diagnosis

DUKE MAJOR CRITERIA

FYI

1. Positive blood cultures

- Typical IE bacteria in two separate cultures separated by time OR
- Persistently positive blood cultures OR
- Single positive culture for *Coxiella burnetii* or anti-phase 1 IgG antibody titer $\geq 1:800$

2. Evidence of endocardial involvement

3. New valvular regurgitation

4. Positive echocardiogram

- TEE or TTE

MINOR CRITERIA

FYI

- Predisposition (e.g. heart condition, IVDU)
- Fever $\geq 38^{\circ}\text{C}$
- Vascular phenomena (e.g. emboli, mycotic aneurysm, ICH, Janeway lesions)
- Immunologic phenomena: Osler nodes, Roth spots, glomerulonephritis, rheumatoid factor
- Microbiological: positive blood culture but does not meet major criteria
- No ECHO criteria (*eliminated in Modified Duke Criteria*)

GOALS OF THERAPY

- ✓ Cure the infection
- ✓ Relieve signs and symptoms
- ✓ Decrease morbidity & mortality associated with the infection
- ✓ Prevent further damage to valves and cardiovascular system
- ✓ Prevent recurrence of IE
- ✓ Educate on IE prophylaxis based on procedure

SURGICAL MEASURES

FYI

- Indicated for life-threatening cases of heart failure secondary to valve damage due to IE
- Other indications: persistent emboli and complications, vegetation >10 mm, antimicrobial treatment failure, severe valvular failure, evidence of extension into surrounding tissues, relapsing infection
- Certain bacteria nearly always require surgery
 - *Brucella spp.*
 - *Candida spp.*

DRUG THERAPY: INTRAVENOUS ANTIBIOTICS

Empiric therapy may be initiated immediately in acutely ill patients with consultation with infectious disease specialist (see point #2)

Obtain blood cultures prior to antimicrobial therapy and use susceptibility data to guide targeted therapy

Duration of therapy starts from 1st day of negative blood cultures and depends on isolated pathogen, patient factors, location and severity of infection

ANTIBIOTIC THERAPY

- Choice of antibiotics depends on
 - Patient risk factors (e.g. PWID, GU procedure)
 - Adherence to regimen
 - Left vs. right-sided IE
 - Native vs. prosthetic valve
 - Pathogen culture, susceptibility and resistance
- Combination antibiotic therapy: concept of synergy with aminoglycosides
- The following guidelines are for culture-positive endocarditis, stratified based on isolated pathogen and its resistance data (MIC used as cut-off points). Treatment options are all to be given intravenously (IV)

TREATMENT GUIDELINES

Staphylococci

Bug	Drug	Native valve	Prosthetic valve
MSSA*	cloxacillin cefazolin	6 weeks (left sided and/or complicated) 2 weeks (right-side and uncomplicated)	6 weeks (beta lactam + rifampin) + 2 weeks of gentamicin
MRSA	vancomycin	6 weeks	6 weeks (vancomycin+ rifampin) + 2 weeks of gentamicin
MRSA or MSSA	daptomycin	6 weeks	6 weeks (daptomycin + rifampin)

*For patients with a beta-lactam (type 1) allergy consider the following: beta-lactam desensitization, cefazolin (patients who are non-anaphylactic to penicillin), or vancomycin

CASE #1

AJ, a 45-year-old male with a history of injection drug use, presents to the emergency department with a 5-day history of fever, chills, and right-sided chest pain that is worsened by inspiration. He is tachycardic and febrile, with a new systolic murmur best heard along the left lower sternal border. Blood cultures grow *Staphylococcus aureus*, and transthoracic echocardiography reveals a vegetation on the tricuspid valve. He is diagnosed with right-sided infective endocarditis, likely related to intravenous drug use. Pulmonary septic emboli are noted on CT chest.

He has a mechanical tricuspid valve.

He had a maculopapular rash to penicillin as a child.

What drug(s), route, and duration would you suggest for the management of this patient's IE?

CASE #2

NJ, a 67-year-old male with a history of hypertension, presents with complaints of progressive fatigue, low-grade fevers, and unintentional weight loss over the past month. He reports mild exertional dyspnea but denies chest pain. On examination, he has a soft diastolic murmur and splinter hemorrhages on his fingernails. Blood cultures grow *Streptococcus gallolyticus* (formerly *S. bovis*), and transesophageal echocardiography shows a vegetation on the aortic valve. A subsequent colonoscopy reveals a colonic adenoma, linking his endocarditis to occult gastrointestinal pathology.

TREATMENT GUIDELINES: NATIVE VALVE IE

Viridans group Streptococci and S. bovis/gallolyticus

Penicillin-susceptible (MIC ≤0.12 mcg/mL)	Relatively penicillin resistant (MIC >0.12 mcg/mL to <0.5 mcg/mL)
Penicillin G x 4 weeks Or Ceftriaxone x 4 weeks Short course: penicillin G or ceftriaxone + gentamicin (only in select low-risk patients) If Beta-lactam allergy: vancomycin x 4 weeks (target trough: 10-15 mcg/mL)	Penicillin G x 4 weeks + gentamicin once daily x 2 weeks If isolate is ceftriaxone susceptible, then ceftriaxone alone may be considered If Beta-lactam allergy: vancomycin x 4 weeks

TREATMENT GUIDELINES: NATIVE VALVE IE

Viridans group Streptococci and S. bovis/gallolyticus

Penicillin-resistant (MIC $\geq$ 0.5 mcg/mL)
Ampicillin or penicillin + gentamicin x 4-6 weeks
Ceftriaxone + gentamicin is a reasonable alternative if susceptible to ceftriaxone x 4-6 weeks
Beta-lactam allergy: vancomycin x 4-6 weeks

Licensed to Shreyansh Salunke (ID: 4477)

TREATMENT GUIDELINES: PROSTHETIC VALVE IE

Viridans group Streptococci and S. bovis/gallolyticus

Penicillin-susceptible (MIC $\leq$ 0.12 mcg/mL)	Relatively penicillin resistant (MIC >0.12 mcg/mL)
(Penicillin G or ceftriaxone) x 6 weeks +/- gentamicin x 2 weeks	(Penicillin G or ceftriaxone) + gentamicin x 6 weeks
Beta-lactam allergy: vancomycin x 6 weeks (target trough: 10-15 mcg/mL)	

CASE #2

NJ, a 67-year-old male with a history of hypertension, presents with complaints of progressive fatigue, low-grade fevers, and unintentional weight loss over the past month. He reports mild exertional dyspnea but denies chest pain. On examination, he has a soft diastolic murmur and splinter hemorrhages on his fingernails. Blood cultures grow *Streptococcus gallolyticus* (formerly *S. bovis*), and transesophageal echocardiography shows a vegetation on the aortic valve. A subsequent colonoscopy reveals a colonic adenoma, linking his endocarditis to occult gastrointestinal pathology.

The MIC for *S. gallolyticus* is 0.25 mg/L.

What other questions do you have?

What recommendations would you have for managing this patient's IE?

TREATMENT GUIDELINES

Enterococci

Penicillin G / ampicillin + gentamicin x 4-6 weeks

Ampicillin + ceftriaxone x 6 weeks (preferred for CrCl <50 mL/min)

B-lactam allergy: vancomycin + gentamicin x 6 weeks

Alternatives if...

- Resistant to gentamicin: above regimens with streptomycin instead of gentamicin
 - Resistant to penicillin: vancomycin + gentamicin
 - Resistant to aminoglycosides, penicillin, and vancomycin: linezolid or daptomycin
-
- Choice of therapy based on susceptibility testing for penicillin/vancomycin (MIC) and high-level resistance to gentamicin
 - Duration of treatment depends on native or prosthetic valve involvement and symptoms

TREATMENT GUIDELINES

Gram negative bacteria: *HACEK group*

3rd generation cephalosporin (ceftriaxone) x 4 weeks if native valve whereas x 6 weeks if prosthetic valve

B-lactam allergy: fluoroquinolone (ciprofloxacin, levofloxacin, moxifloxacin) and consult ID specialist

Avoid ampicillin/penicillin (resistance) and gentamicin (nephrotoxicity)

TREATMENT GUIDELINES: FUNGAL AND CULTURE NEGATIVE IE

Consult with ID specialist

Valve surgery should be done in most cases of fungal IE

Life-long oral azole therapy may be needed for fungal IE

Avoid gentamicin

CASE #3

AK, a 72-year-old male with a history of benign prostatic hyperplasia and chronic kidney disease, presents with fatigue, intermittent fevers, and new-onset confusion over the past two months. He underwent cystoscopy for evaluation of hematuria approximately 3 months ago. Upon examination, he is febrile and has a new systolic murmur at the apex. Laboratory work reveals leukocytosis and mild renal dysfunction. Blood cultures grow *gram-positive cocci in pairs and short chains* in multiple sets. Transesophageal echocardiography shows a vegetation on the mitral valve. Given his recent urologic procedure, age and echocardiography findings, the diagnosis of native valve infective endocarditis is made, and targeted antimicrobial therapy is initiated in consultation with Infectious Diseases. He has no known allergies

What is the most likely pathogen in the patient?

What empiric therapy would you suggest pending speciation and susceptibility testing?

PHARMACOTHERAPEUTIC OPTIONS

What about oral antimicrobials?

- POET Trial (NEJM 2019)
- Combination of oral antimicrobials after initial IV therapy
- Non-inferior to IV therapy
- Endorsed by most current IE guidelines (European)
- **Challenges:**
 - Different regimes than we typically use
 - MRSA not common in study population
 - Adherence with high burden drug therapy difficult
- **Opportunities:**
 - Fewer days in hospital
 - Fewer line-related complications
 - Potentially better outcomes

IE ANTIBIOTIC PROPHYLAXIS

- Controversial due to lack of evidence
- Associated with fewer benefits and greater adverse events
- Antibiotic coverage for VGS during procedure and few hours after
- If patient is already on antibiotic used for IE prophylaxis, choose agent from different class OR delay procedure until 10+ days after course completion to allow flora to return to natural state
- Revised AHA/ACC recommendations limits prophylaxis ONLY to high-risk individuals undergoing select dental, GI, GU tract procedures
- High-risk patient groups (per updated 2017 AHA/ACC IE prophylaxis guidelines):
 - Prosthetic cardiac valves or prosthetic material used for cardiac valve repair
 - Previous IE
 - Unrepaired cyanotic congenital heart disease (CHD)
 - Repaired CHD with residual shunts or valvular regurgitation
 - Cardiac transplant recipients with valve regurgitation

PROCEDURES REQUIRING PROPHYLAXIS

- **Dental:** manipulation of gingival tissue or the periapical region of teeth or perforation of the oral mucosa
 - Not indicated for dental radiographs, orthodontic placement/adjustment, shedding of deciduous teeth, and bleeding from trauma to lips or oral mucosa
- **Respiratory:** incision or biopsy of respiratory mucosa (e.g. tonsillectomy)
- **Infected skin, skin structures or musculoskeletal tissue**

PROPHYLAXIS

Situation	Drug	Adult	Child
Oral	Amoxicillin	2 g PO	50 mg/kg
Patient who cannot take medication orally	Ampicillin	2 g IM/IV	50 mg/kg IM/IV
	or Cefazolin/ceftriaxone	1 g IM/IV	50 mg/kg IM/IV
Penicillin Allergy (Oral)	Cephalexin*† (non-severe, non-type 1 allergy)	2g PO	50 mg/kg
	or Azithromycin or clarithromycin	500 mg PO	15 mg/kg
	or Doxycycline	100 mg	<45 kg, 2.2 mg/kg >45 kg, 100 mg
Penicillin Allergy (IV/IM)	Cefazolin/ceftriaxone†	1g IM or IV	50 mg/kg IM/IV

- Give 30-60 min prior to procedure
- If prophylaxis is missed before procedure, it can be given up to 2 hours after the procedure

* Or other first- or second-generation oral cephalosporin in equivalent adult or pediatric dosing

† Cephalosporins should not be used in patients with a history of anaphylaxis, angioedema, or urticaria with penicillin or ampicillin

MONITORING

Parameter	Frequency
Chills	Daily until resolution
Fever	Daily until afebrile
Other presenting clinical symptoms	Daily during course of treatment
Blood culture	Q24-48 hours until culture negative
Blood Glucose	Daily until normalized
WBC	Daily until WBC are normalized
Creatinine Clearance	Twice weekly for duration of therapy
ECHO	Repeat or follow-up if needed
Side effects from therapy (gentamicin, vancomycin)	Daily during course of treatment
Therapeutic Drug Monitoring To achieve appropriate levels (gentamicin, vancomycin)	After 3 rd dose (pre & post) then weekly More frequently (q2-3d) if renal function decreases

TAKEAWAY/THERAPEUTIC TIPS

- Use patient risk factors to identify etiology
- Patient with isolated *S. bovis/gallolyticus* IE should have colonoscopy to investigate for colon cancer
- Treat with bactericidal antibiotics
- Complete TDM (therapeutic drug monitoring) for aminoglycoside and vancomycin to reduce adverse reactions and improve efficacy
- Duration of antimicrobial therapy should begin on the first day on which blood cultures are negative (if surgical cultures negative)

CASE

IE is a 62-year-old female presenting to your emergency department with night sweats, chills and fatigue. IE was in her usual state of health until approximately 3 weeks ago. At this time, she noted subjective fevers and felt generally unwell. Over the next several days, she slept poorly, had night sweats and recurring chills. She had a single documented fever of 38.1 °C approximately a week prior to presentation. Two weeks ago, prior to symptom onset, she had a root canal surgery. She does not have any allergies to any medications, and she does not take any medications. She lives with her partner, works as a first-grade teacher, and is an avid gardener. She does not drink alcohol or smoke.

Course

- IE has blood cultures drawn in the ED that return as gram-positive cocci in chains. 3 of 3 blood cultures separated by an hour
- These cultures return positive for Viridans-group Streptococci
- Her transthoracic echo reveals a large vegetation on the aortic valve



CASE - QUESTIONS

1. Which of the following is the most important first step in the management of IE?
 - a) Repeat blood cultures
 - b) Perform transesophageal echocardiography
 - c) Obtain CV surgery consult
 - d) Start antimicrobials

2. On day 3 of hospital admission, IE's culture results return as Viridans group streptococci with a penicillin MIC of 0.09 mcg/mL. What is the optimal management of her antimicrobial therapy at this point?
 - a) Ampicillin 2 g IV Q4H x 6 weeks
 - b) Penicillin G 2 MU IV Q4H x 6 weeks
 - c) Ceftriaxone 2 g IV Q24H x 4 weeks
 - d) Vancomycin 15 mg/kg IV Q12H x 4 weeks

CASE - QUESTIONS

3. What is the best monitoring parameter to support clinical resolution for IE?
- a) CRP
 - b) SCr
 - c) Blood culture
 - d) Follow up TTE
4. Which safety parameter is most important to include for antimicrobial therapy?
- a) Visual acuity testing
 - b) EEG
 - c) ALT
 - d) SCr

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CHANGE LOG

March 2025

- Slide 19: Added information for MSSA beta lactam allergy recommendations
- Slide 23: Added more information about alternatives for resistant cases

July 2025

- Formatting changes made throughout
- Slide 19: Updated regimen for MRSA in prosthetic valve
- Added slides 4, 20, 21, 25, and 29 – Case studies
- Slides 37-38: Updated case practice questions

October 2025

- Slide 33: Updated agents recommended for prophylaxis in penicillin allergy (removed clindamycin, added doxycycline, and added footnotes)
- Slide 39: Updated references

February 2026

- Slide 2: Updated NAPRA competencies
- Minor formatting edits; no content changes made